

Soham Deshpande

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Objective

AI & ML postgraduate with hands-on experience across computer vision, generative AI, and full-stack development. Built and evaluated deep learning models (GANs, object detection) with rigorous ablation studies and quantitative metrics, developed real-time AI applications using multiple LLM/API providers, and shipped full-stack web applications with REST APIs and relational databases. Comfortable working independently, reading documentation, and learning new tools and domains quickly.

Projects

1. InfraNova AI — Thermal Infrared to RGB Satellite Image Translation (PyTorch, GANs, Computer Vision, FastAPI)

- Developed a cross-spectral satellite image translation system using a Pix2PixHD-style Global ResNet GAN to synthesize RGB imagery from Landsat 9 TIRS-2 Band 10 + Band 11 inputs.
- Built and evaluated an 11.37M-parameter generator using LSGAN and multi-term loss optimization, achieving 13.74 dB PSNR, 0.4502 SSIM, and 0.3711 YOLOv8 F1@0.25 on 1,259 held-out samples.
- Engineered a FastAPI + React/Vite application with tiled whole-raster inference (128×128 tiles, cosine/Hann blending), processing a 1002×1001 scene in 1.74s at 69.5 tiles/sec, plus TTA and CLAHE post-processing.
- Conducted controlled ablation experiments (Exp1–Exp4) with strict single-variable changes and built a reproducible Kaggle GPU training workflow with environment fingerprinting, SHA-256 artifact verification.

2. Translytic — AI-Based Real-Time Video Captioning & Translation System (Python, Whisper, Gemini API)

- Expanded a real-time video captioning and translation system using Python, OpenAI Whisper, Gemini API, Google Translate API, edge-tts, OpenCV, and Tkinter.
- Built an end-to-end AI pipeline for speech-to-text, translation, subtitle generation, and multilingual dubbing with timestamp synchronization.
- Integrated REST APIs and JSON-based data processing with automatic fallback mechanisms for transcription and translation.
- Applied multithreaded video playback, frame buffering, and audio-video synchronization for smooth real-time performance.

3. NoteFlow — Full Stack Notes Management Platform (Django, MySQL, REST APIs, JavaScript)

- Improved a full-stack notes management application using Django, MySQL, HTML, CSS, and JavaScript.
- Enforced user authentication, CRUD operations, category-based filtering, and role-based access control.
- Designed relational database schemas using Django ORM and MySQL with support for per-user notes and dashboard analytics.
- Built AJAX-based pin functionality and real-time search capabilities without full page refreshes.

Education

Master of Computer Applications (AI & ML)
Ramdeobaba University, Nagpur | 2025 – Present

Bachelor of Computer Applications
G.H. Raisoni College of Engineering and Management, Nagpur | 2022 – 2025

Skills

Technical Skills

- Computer Vision & Deep Learning: PyTorch, OpenCV, GANs (Pix2PixHD), YOLOv8, object detection, image-to-image translation, satellite/thermal imagery processing
- Generative AI & LLMs: OpenAI Whisper, Gemini API, ChatGPT, Claude, prompt design, API-based AI integration
- Programming Languages: Python, Java, SQL
- Data & Pipelines: Image preprocessing, normalization, data validation, reproducible training pipelines, GPU training (multi-GPU, AMP)
- Testing & Validation: Pytest, structured experiment validation, ablation studies, pipeline evaluation
- Tools & Collaboration: Git, GitHub, technical documentation, cross-functional teamwork
- Backend & Full-Stack: Django, FastAPI, REST APIs, MySQL, HTML, CSS, JavaScript, NumPy, Pandas, Power BI, Excel

Soft Skills:

Analytical Thinking | Research & Documentation | Problem Solving | Communication | Team Collaboration

Certifications

- Google Cloud Career Launchpad – Computing Foundations Track, Google Cloud (Apr 2026)
- Assets, Threats, and Vulnerabilities – Google Cybersecurity Certificate, Coursera (Apr 2026)
- Connect and Protect: Networks and Network Security – Google Cybersecurity Certificate, Coursera (Mar 2026)
- Tools of the Trade: Linux and SQL – Google Cybersecurity Certificate, Coursera (Apr 2026)
- Fundamentals of Deep Learning – NVIDIA Certificate of Competency (Aug 2026)